

SentinelCleanAI: Autonomous Edge Vision Framework for Intelligent Robotic Maintenance

Cuong Dang (Individual Project)

Tier 2 (Advanced - Custom Fine-Tuning & Local Robotics Orchestration)

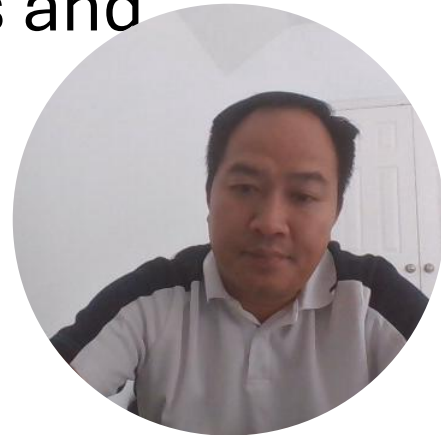
Repo Link: <https://github.com/binxixi23/SentinelCleanAIFN>

Video Link: <https://www.youtube.com/watch?v=fHtxf9PLgRU>



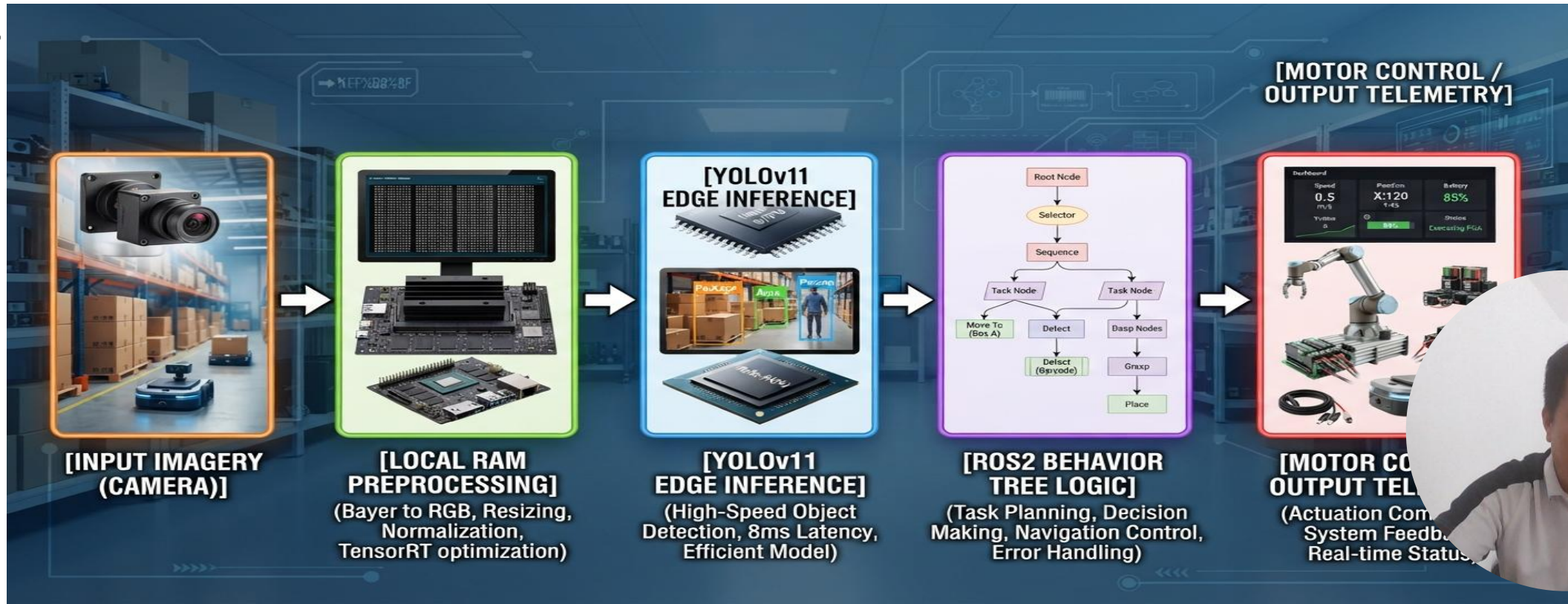
The Problem

- The Industrial Blindspot: Heavy reliance on raw proximity/distance sensors
- The Critical Failure Point: Inability to distinguish dry debris from liquid spills
- The Financial Cost: Vacuuming over fluids instantly destroys internal paper machinery filters
- The Maintenance Damage: Ruined electric suction motors and cross-contamination of clean floor spaces



What You Built

- The Architecture: Privacy-First Vision Core + Low-Level Motor Control
- Onboard Data Minimization: Processing locks entirely within local volatile RAM buffers



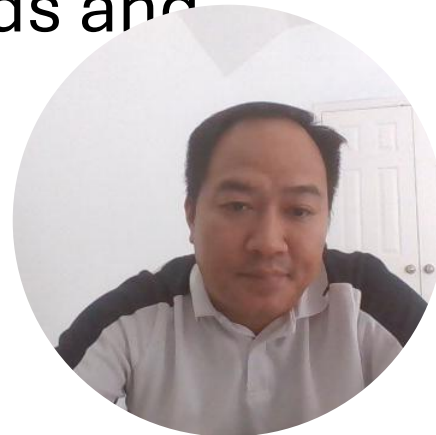
System Demo

- Functional Proof of Concept Deployed and Active
- Onboard Edge Execution: Instantaneous boundary-box mapping on unseen floor frames
- Hardware Fail-Safe Strategy: Meta SAM 2 segmenter triggers *conditionally* on active spill keys
- Production Media Backup: Live execution logs archived on unlisted YouTube channel



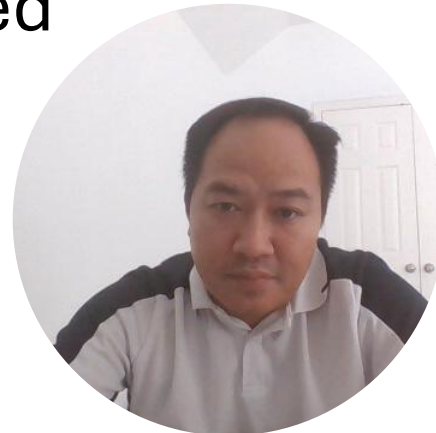
How It Works

- Primary Detector Object Core: Ultralytics YOLOv11 nano (yolo11n.pt)
- Target Validation Specialist: Meta SAM 2 (Zero-Shot Pixel Segmentation Core)
- Local Microcontroller Compliance: Optimized layers running at standard 640x640 resolution scale
- Zero Cloud Dependencies: Eliminates API latency overloads and internet bandwidth risks

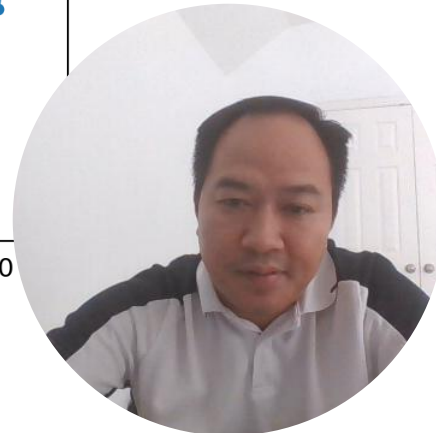
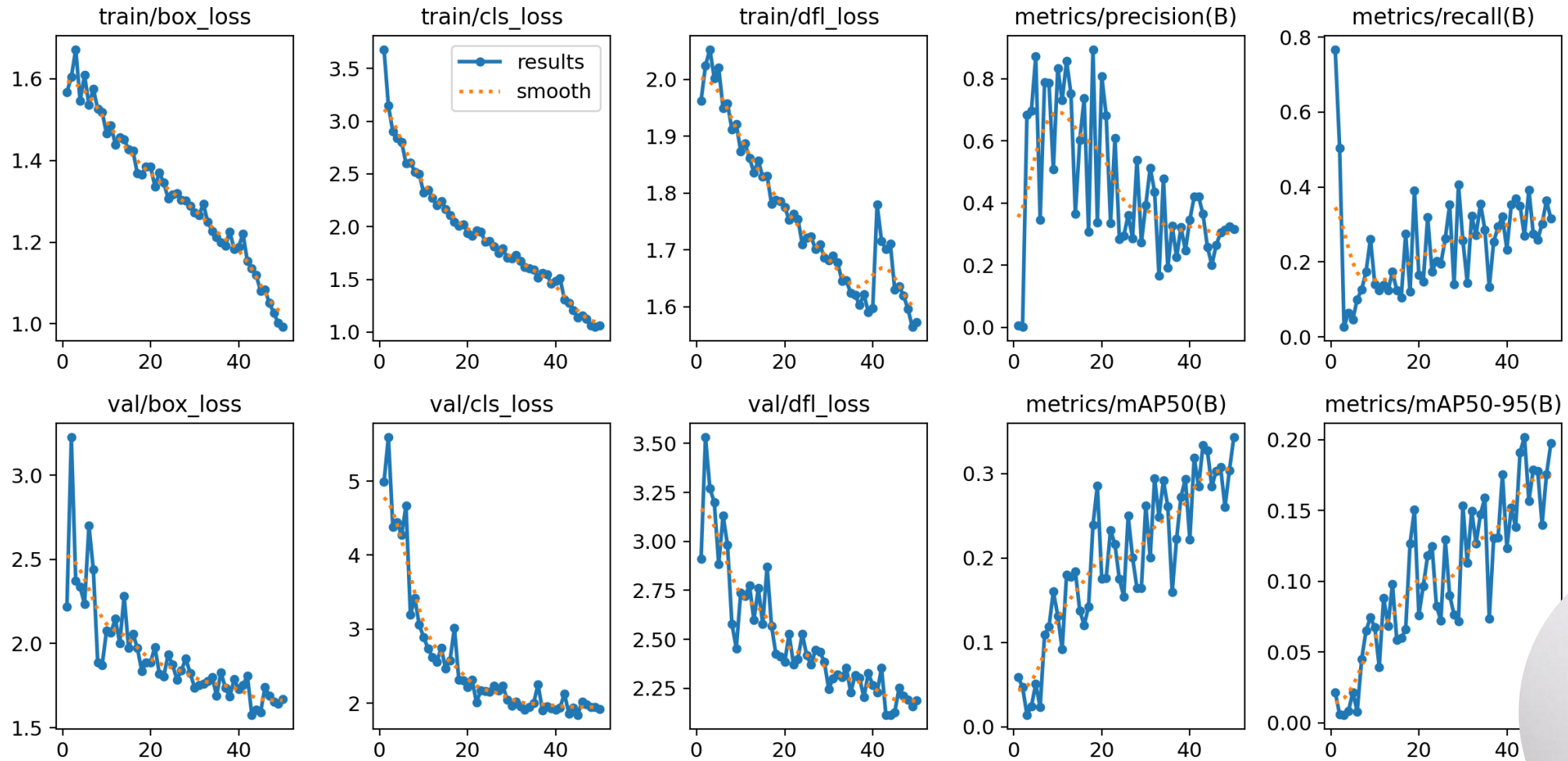


The Data You Used

- Source Repository: Verified public benchmark profiles via Roboflow Universe (ntut-fkont/dirty-detect)
- Dataset Scale: 900 Training Images | 69 Validation Images (Strict folder separation)
- Rigid Annotation Cleaning: Hand-scrubbed images to clear out ambiguous ambient room lighting noise
- Evaluated Multiclass Array: Water Stains, Brown Stains, Red Stains, Tea Stains, White Stains, Other



Results vs Your Targets

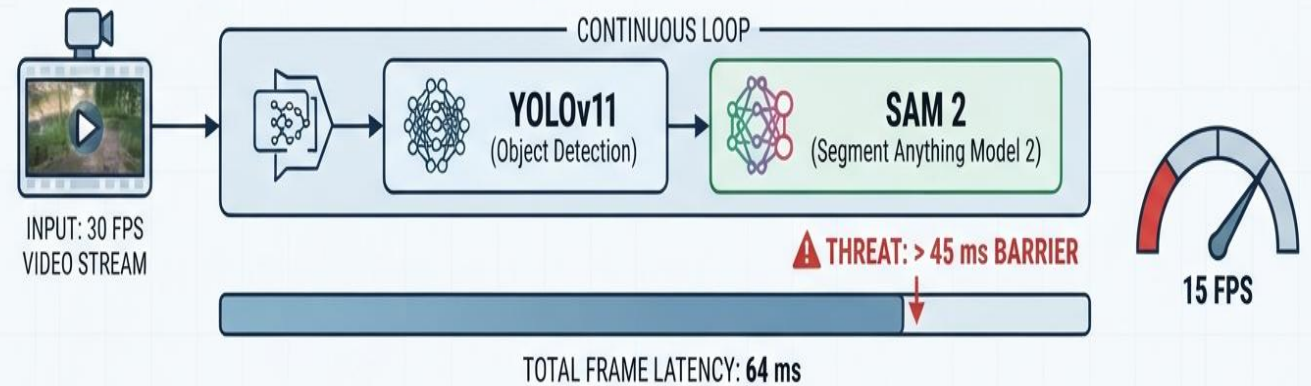


The Architectural Fail-Safe Paradigm (Plan B Implementation)

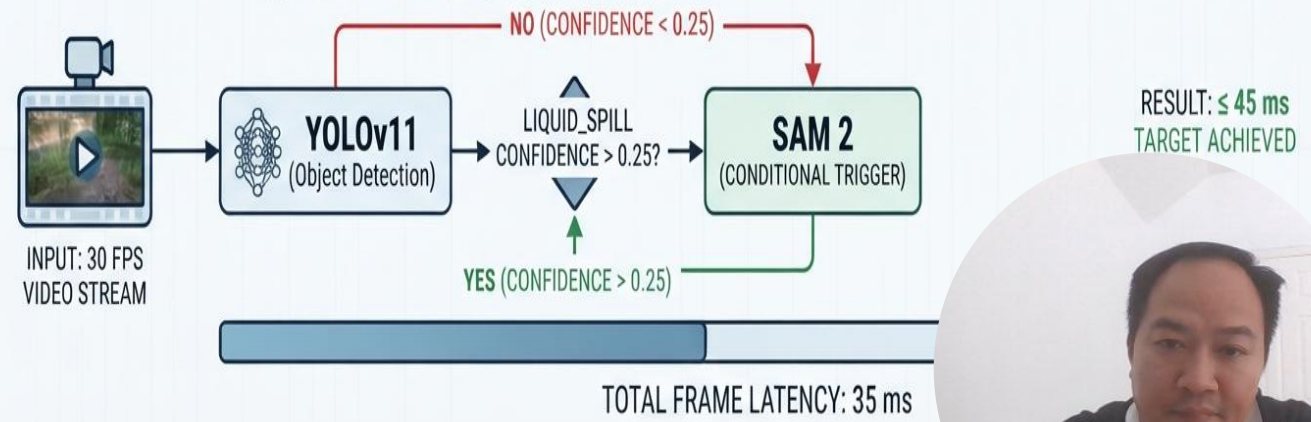
- The Latency Countermeasure: Hard-locking primary tracker tracking threshold to a stable 0.25 confidence index
- The Processing Governor: Conditionally initializing Meta SAM 2 *only* when fluid keys pass confidence checks
- The Zero-Light Contingency: Low-level navigation fallback to active onboard Infrared (IR) Time-of-Flight range tracking

THE TECHNICAL PIVOT: RECOVERING FRAME SPEED

BEFORE: Continuous SAM 2 (Overhead Threat)

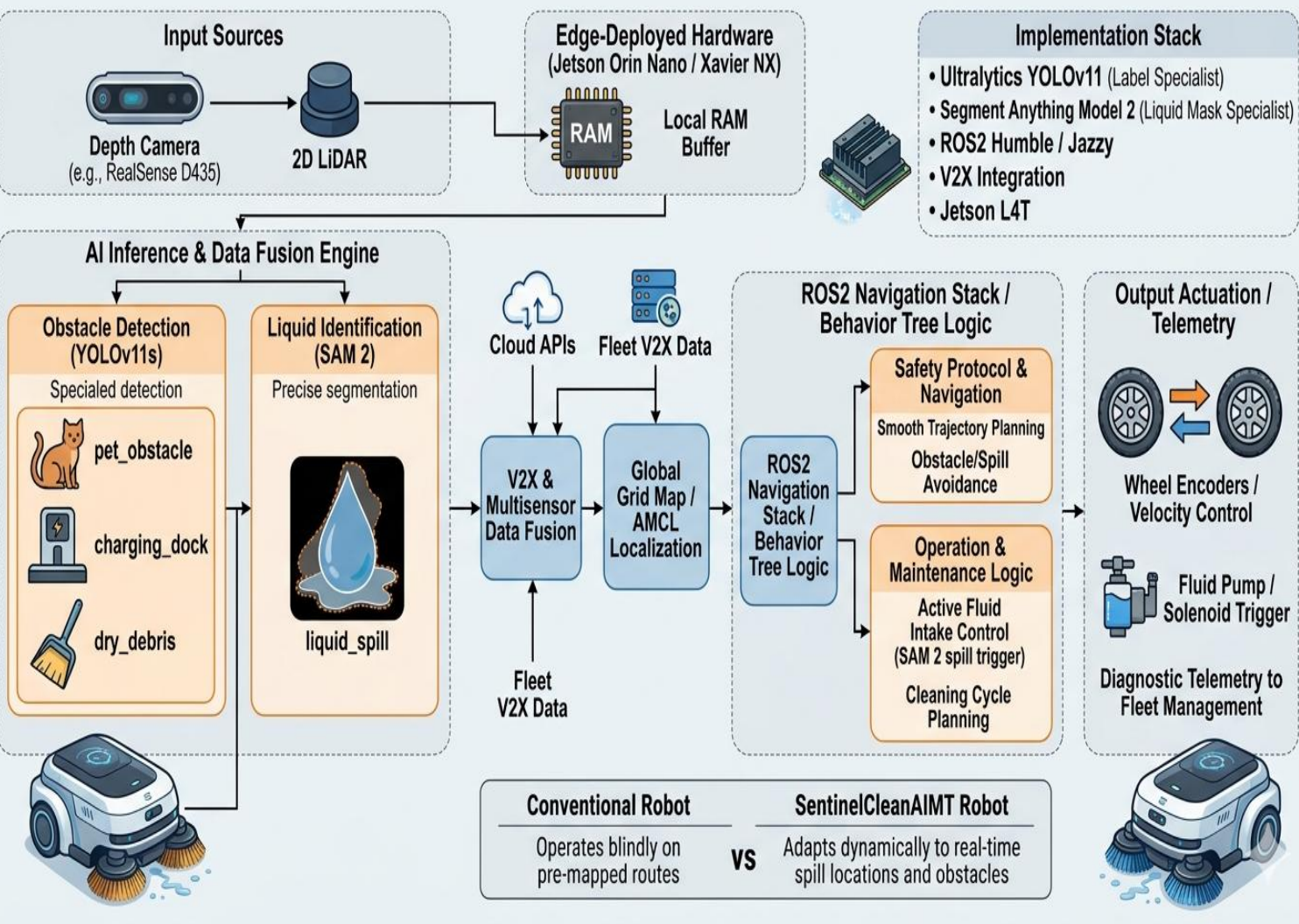


AFTER: Conditional Trigger (Plan B Deployed)



- The grid contains 16 images of stains, each with a title and a bounding box. The titles are: 'Water Stains0.5', 'Water Stains0.8', 'Water Stains0.7', 'Water Stains0.6', 'Brown Stains0.3', 'Water Stains0.9', 'Water Stains0.6', 'Brown Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3', 'Water Stains0.3'. The bounding boxes are: [102, 102, 245, 245], [245, 102, 388, 245], [388, 102, 531, 245], [531, 102, 674, 245], [674, 102, 817, 245], [817, 102, 960, 245], [102, 245, 245, 388], [245, 245, 388, 388], [388, 245, 531, 388], [531, 245, 674, 388], [674, 245, 817, 388], [817, 245, 960, 388], [102, 388, 245, 531], [245, 388, 388, 531], [388, 388, 531, 531], [531, 388, 674, 531], [674, 388, 817, 531], [817, 388, 960, 531], [102, 531, 245, 674], [245, 531, 388, 674], [388, 531, 531, 674], [531, 531, 674, 674], [674, 531, 817, 674], [817, 531, 960, 674], [102, 674, 245, 817], [245, 674, 388, 817], [388, 674, 531, 817], [531, 674, 674, 817], [674, 674, 817, 817], [817, 674, 960, 817], [102, 817, 245, 960], [245, 817, 388, 960], [388, 817, 531, 960], [531, 817, 674, 960], [674, 817, 817, 960], [817, 817, 960, 960].

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Lessons and Next Steps

- The Engineering Lesson: High-quality Data-Centric AI curation outperforms excessive hyperparameter tuning
- The Architectural Lesson: Edge deployment successfully guarantees 100% data minimization and privacy
- Future Milestones: Moving layers toward NMS-F... Maximum Suppression to-end vision matrices



Final Failure Case Analysis

- The Open Failure Vulnerability: Low-contrast clear liquid puddles on flat, white floor tiles
- Why Neural Networks Struggle: Faint fluid borders blend entirely with background pixel textures
- System Safety Overwrite: Low-level ROS2 navigational steering bypass logic
- If edge model tracking confidence falls below 0.25, the unit executes an automatic 90-degree escape turn

